

T3 Lagrangian and Weinberg-operator report

EFT order: 5; loop order: 1.

$$T3 - B, \alpha = -1 : \quad S_1 = (2, -\frac{1}{2}), \quad S_2 = (2, \frac{1}{2}), \quad F = (1, 0)$$

Build T3 ingredients	Success Success	Matching Weinberg extraction	Success Success
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Accepted interactions

Yukawa1_GenericCG, Yukawa2_GenericCG, Scalar1SelfInv1, Scalar2SelfInv1, HiggsPortal1Inv1, HiggsPortal1Inv2, HiggsPortal2Inv1, HiggsPortal2Inv2, ScalarCrossPortalInv1, ScalarCrossPortalInv2, T3ScalarMix_GenericCG.

BSM free Lagrangian $\mathcal{L}_{\text{free}}$

$$\begin{aligned} & \frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1}) \\ & - \frac{1}{2} M_F N_{r_1}^T C N_{r_1} \\ & + D_{\mu_1} (\text{NewScalar1}_{j_1}) D_{\mu_1} (\text{NewScalar1}_{j_1})^\dagger \\ & + D_{\mu_1} (\text{NewScalar2}_{j_1}) D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger \\ & + M_{S1}^2 (-\text{NewScalar1}_{j_1}) \text{NewScalar1}_{j_1}^\dagger \\ & - M_{S2}^2 \text{NewScalar2}_{j_1} \text{NewScalar2}_{j_1}^\dagger \end{aligned}$$

Interaction Lagrangian $\mathcal{L}_{\text{int}}$

$$\begin{aligned} & \overline{\ell_{a_1, i_1}} P_R N_{r_1} \text{NewScalar1}_{j_1} \text{T3Y1CG}_{a_1, j_1} y_{1, i_1, r_1} \\ & + \overline{y_{2, i_1, r_1}} N_{r_1}^T C P_L \ell_{a_1, i_1} \text{NewScalar2}_{j_1} \overline{\text{T3Y2CG}_{a_1, j_1}} \\ & + \overline{y_{1, i_1, r_1}} N_{r_1}^T C P_L \ell_{a_1, i_1} \overline{\text{T3Y1CG}_{a_1, j_1}} \text{NewScalar1}_{j_1}^\dagger \\ & + \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \overline{\lambda_{T3}} \text{NewScalar2}_{j_2} \overline{\text{T3MixCG}_{a_1, a_2, j_1, j_2}} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\ & + \overline{\ell_{a_1, i_1}} P_R N_{r_1} \text{T3Y2CG}_{a_1, j_1} y_{2, i_1, r_1} \text{NewScalar2}_{j_1}^\dagger \\ & + H_{a_1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3MixCG}_{a_1, a_2, j_2, j_1} \lambda_{T3} \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambda12Inv1} \text{NewScalar1}_{j_3} \text{NewScalar2}_{j_4} \text{T3CrossScalarCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \text{lambda12Inv2} \text{NewScalar1}_{j_3} \text{NewScalar2}_{j_4} \text{T3CrossScalarCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \end{aligned}$$

BSM-induced EFT contribution $\Delta\mathcal{L}_{\text{EFT}}$

$$\begin{aligned} & - \frac{\hbar \log \left(\frac{\mu_{\text{bar2}}}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S1}^2)^2} \\ & - \frac{\hbar \log \left(\frac{\mu_{\text{bar2}}}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S2}^2)^2} \\ & - \frac{\hbar \log \left(\frac{\mu_{\text{bar2}}}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S1}^2)^2} \end{aligned}$$

$$\begin{aligned}
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{12\epsilon} \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu \text{bar}^2}{M_{S1}^2}\right) B_{\mu_1 \mu_2}^2 \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu \text{bar}^2}{M_{S2}^2}\right) B_{\mu_1 \mu_2}^2 \\
& - \frac{\hbar (W_{\mu_1 \mu_2}^{A_1})^2}{12\epsilon} \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu \text{bar}^2}{M_{S1}^2}\right) (W_{\mu_1 \mu_2}^{A_1})^2 \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu \text{bar}^2}{M_{S2}^2}\right) (W_{\mu_1 \mu_2}^{A_1})^2 \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu \text{bar}^2}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu \text{bar}^2}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu \text{bar}^2}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu \text{bar}^2}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S2}^2)^2} \\
& + C_{H2} H_{a_1} H_{a_1}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{\text{lambdaH1Inv}1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{5 \text{lambdaH1Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& + \frac{\text{lambdaH2Inv}1^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{5 \text{lambdaH2Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& + \frac{\text{lambdaH1Inv}1 \text{lambdaH1Inv}2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{\text{lambdaH2Inv}1 \text{lambdaH2Inv}2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{1}{2} \text{lambdaH1Inv}1^2 \hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6} \text{lambdaH1Inv}2^2 \hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH1Inv}1 \text{lambdaH1Inv}2 \hbar \log\left(\frac{\mu_{S1}^{\text{bar}2}}{M_{S1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{1}{2} \text{lambdaH2Inv}1^2 \hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{5}{6} \text{lambdaH2Inv}2^2 \hbar \log\left(\frac{\mu_{S2}^{\text{bar}2}}{M_{S2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \frac{\text{lambdaH2Inv1lambdaH2Inv2}\hbar \log\left(\frac{\mu_{\text{bar2}}}{M_{\text{S2}}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& + \frac{4\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon} \\
& + \frac{4\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S1}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{\text{S1}}^2 - M_{\text{S2}}^2)} \\
& + \frac{4\hbar \log\left(\frac{\mu_{\text{bar2}}}{M_{\text{S1}}^2}\right) \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S1}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{\text{S1}}^2 - M_{\text{S2}}^2)} \\
& - \frac{4\hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S2}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{\text{S1}}^2 - M_{\text{S2}}^2)} \\
& - \frac{4\hbar \log\left(\frac{\mu_{\text{bar2}}}{M_{\text{S2}}^2}\right) \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{\text{S2}}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{\text{S1}}^2 - M_{\text{S2}}^2)}
\end{aligned}$$

Weinberg sector

$$- \frac{2\hbar M_F \left(M_F^2 \left(M_{\text{S1}}^2 \log\left(\frac{M_F^2}{M_{\text{S1}}^2}\right) - M_{\text{S2}}^2 \log\left(\frac{M_F^2}{M_{\text{S2}}^2}\right) \right) + M_{\text{S1}}^2 M_{\text{S2}}^2 \left(\log\left(\frac{M_F^2}{M_{\text{S2}}^2}\right) - \log\left(\frac{M_F^2}{M_{\text{S1}}^2}\right) \right) \right) \left(\lambda_{\text{T3}} H_{a_1} H_{a_3} \bar{\epsilon}_{a_1, a_2} \overline{y_{1, i_1, r_1}} \bar{\epsilon}_{a_3, a_4} \overline{y_{2, i_2, r_1}} \ell_{a_4, i_2}{}^T C P_L \ell_{a_2, i_1} + \overline{\lambda_{\text{T3}}} \epsilon_{a_1, a_3} y_{1, i_1, r_1} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger y_{2, i_2, r_1} \overline{\ell_{a_3, i_1}} C P_R \overline{\ell_{a_4, i_2}}{}^T \right)}{\sqrt{3} (M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)}$$

Weinberg coefficient C_5

$$- \frac{2}{3} \left(- \frac{\sqrt{3}\hbar M_F^3 M_{\text{S2}}^2 \lambda_{\text{T3}} \log\left(\frac{M_F^2}{M_{\text{S2}}^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1}}{(M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)} + \frac{\sqrt{3}\hbar M_F^3 M_{\text{S1}}^2 \lambda_{\text{T3}} \log\left(\frac{M_F^2}{M_{\text{S1}}^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1}}{(M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)} - \frac{\sqrt{3}\hbar M_F M_{\text{S1}}^2 M_{\text{S2}}^2 \lambda_{\text{T3}} \log\left(\frac{M_F^2}{M_{\text{S1}}^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1}}{(M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)} + \frac{\sqrt{3}\hbar M_F M_{\text{S1}}^2 M_{\text{S2}}^2 \lambda_{\text{T3}} \log\left(\frac{M_F^2}{M_{\text{S2}}^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1}}{(M_F^2 - M_{\text{S1}}^2) (M_F^2 - M_{\text{S2}}^2) (M_{\text{S1}}^2 - M_{\text{S2}}^2)} \right)$$

$$T3 - C, \alpha = -1: \quad S_1 = (2, -\frac{1}{2}), \quad S_2 = (2, \frac{1}{2}), \quad F = (3, 0)$$

Build T3 ingredients	Success Success	Matching Weinberg extraction	Success Success
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Accepted interactions

Yukawa1_GenericCG, Yukawa2_GenericCG, Scalar1SelfInv1, Scalar2SelfInv1, HiggsPortal1Inv1, HiggsPortal1Inv2, HiggsPortal2Inv1, HiggsPortal2Inv2, ScalarCrossPortalInv1, ScalarCrossPortalInv2, T3ScalarMix_GenericCG.

BSM free Lagrangian $\mathcal{L}_{\text{free}}$

$$\begin{aligned} & \frac{1}{2} i N_{j_1, r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{j_1, r_1}) \\ & - \frac{1}{2} M_F N_{j_1, r_1}^T C N_{j_1, r_1} \\ & + D_{\mu_1} (\text{NewScalar1}_{j_1}) D_{\mu_1} (\text{NewScalar1}_{j_1})^\dagger \\ & + D_{\mu_1} (\text{NewScalar2}_{j_1}) D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger \\ & + M_{S1}^2 (-\text{NewScalar1}_{j_1}) \text{NewScalar1}_{j_1}^\dagger \\ & - M_{S2}^2 \text{NewScalar2}_{j_1} \text{NewScalar2}_{j_1}^\dagger \end{aligned}$$

Interaction Lagrangian $\mathcal{L}_{\text{int}}$

$$\begin{aligned} & \overline{\ell_{a_1, i_1}} P_R N_{j_1, r_1} \text{NewScalar1}_{j_1} \text{T3Y1CG}_{a_1, j_1, j_1} y_{1, i_1, r_1} \\ & + \overline{y_{2, i_1, r_1}} N_{j_1, r_1}^T C P_L \ell_{a_1, i_1} \text{NewScalar2}_{j_1} \text{T3Y2CG}_{a_1, j_1, j_1} \\ & + \overline{y_{1, i_1, r_1}} N_{j_1, r_1}^T C P_L \ell_{a_1, i_1} \text{T3Y1CG}_{a_1, j_1, j_1} \text{NewScalar1}_{j_1}^\dagger \\ & + \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \overline{\lambda_{\text{T3}}} \text{NewScalar2}_{j_2} \text{T3MixCG}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\ & + \overline{\ell_{a_1, i_1}} P_R N_{j_1, r_1} \text{T3Y2CG}_{a_1, j_1, j_1} y_{2, i_1, r_1} \text{NewScalar2}_{j_1}^\dagger \\ & + H_{a_1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3MixCG}_{a_1, a_2, j_2, j_1} \lambda_{\text{T3}} \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambda12Inv1} \text{NewScalar1}_{j_3} \text{NewScalar2}_{j_4} \text{T3CrossScalarCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \text{lambda12Inv2} \text{NewScalar1}_{j_3} \text{NewScalar2}_{j_4} \text{T3CrossScalarCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \end{aligned}$$

BSM-induced EFT contribution $\Delta\mathcal{L}_{\text{EFT}}$

$$\begin{aligned} & - \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S1}^2)^2} \\ & - \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S2}^2)^2} \\ & - \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S1}^2)^2} \\ & - \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{2, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S2}^2)^2} \end{aligned}$$

$$\begin{aligned}
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F^3}{3(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F^3}{3(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{3(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{3(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S2}^2)^2} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S2}^2)^2} \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S1}^2)} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S1}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{3}\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S2}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4(M_F^2 - M_{S2}^2)^2} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S2}^2)^2} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{3(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)}
\end{aligned}$$

$$\begin{aligned}
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} \ell_{a_4,i_2}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{3(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{3(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{3(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{12\epsilon} \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) B_{\mu_1 \mu_2}^2 \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) B_{\mu_1 \mu_2}^2 \\
& - \frac{13\hbar (W_{\mu_1 \mu_2}^{A_1})^2}{12\epsilon} \\
& - \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_F^2}\right) (W_{\mu_1 \mu_2}^{A_1})^2 \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) (W_{\mu_1 \mu_2}^{A_1})^2 \\
& - \frac{1}{24} \hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) (W_{\mu_1 \mu_2}^{A_1})^2 \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \sqrt{3}\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \sqrt{3}\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S2}^2)^2} \\
& + C_{H2} H_{a_1} H_{a_1}^\dagger \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{8}\sqrt{3\hbar y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger \\
& -\frac{\sqrt{3\hbar y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{2(M_F^2-M_{S1}^2)} \\
& -\frac{\sqrt{3\hbar}\log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^2y_{1,i_3,r_1}H_{a_1}^\dagger}{2(M_F^2-M_{S1}^2)} \\
& -\frac{\sqrt{3\hbar y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{4(M_F^2-M_{S1}^2)^2} \\
& -\frac{\sqrt{3\hbar}\log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S1}^4y_{1,i_3,r_1}H_{a_1}^\dagger}{4(M_F^2-M_{S1}^2)^2} \\
& +\frac{\sqrt{3\hbar y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon} \\
& -\frac{1}{8}\sqrt{3\hbar y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger \\
& -\frac{\sqrt{3\hbar y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S2}^2y_{2,i_3,r_1}H_{a_1}^\dagger}{2(M_F^2-M_{S2}^2)} \\
& -\frac{\sqrt{3\hbar}\log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S2}^2y_{2,i_3,r_1}H_{a_1}^\dagger}{2(M_F^2-M_{S2}^2)} \\
& -\frac{\sqrt{3\hbar y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S2}^4y_{2,i_3,r_1}H_{a_1}^\dagger}{4(M_F^2-M_{S2}^2)^2} \\
& -\frac{\sqrt{3\hbar}\log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}e_{i_2}}P_L\ell_{a_1,i_1}M_{S2}^4y_{2,i_3,r_1}H_{a_1}^\dagger}{4(M_F^2-M_{S2}^2)^2} \\
& -H_{a_1}\mu^2H_{a_1}^\dagger \\
& +\frac{\text{lambdaH1Inv1}^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{5\text{lambdaH1Inv2}^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& +\frac{\text{lambdaH2Inv1}^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& +\frac{5\text{lambdaH2Inv2}^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon} \\
& +\frac{\text{lambdaH1Inv1}\text{lambdaH1Inv2}\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& +\frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& +\frac{1}{2}\text{lambdaH1Inv1}^2\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{5}{6}\text{lambdaH1Inv2}^2\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{\text{lambdaH1Inv1}\text{lambdaH1Inv2}\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}} \\
& +\frac{1}{2}\text{lambdaH2Inv1}^2\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{5}{6}\text{lambdaH2Inv2}^2\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger \\
& +\frac{\text{lambdaH2Inv1}\text{lambdaH2Inv2}\hbar\log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}}
\end{aligned}$$

$$\begin{aligned}
& + \frac{4\hbar\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon} \\
& + \frac{4\hbar\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}M_{\text{S}1}^2\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& + \frac{4\hbar\log\left(\frac{\mu_{\text{S}1}^{\text{bar}2}}{M_{\text{S}1}^2}\right)\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}M_{\text{S}1}^2\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& - \frac{4\hbar\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}M_{\text{S}2}^2\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{\text{S}1}^2 - M_{\text{S}2}^2)} \\
& - \frac{4\hbar\log\left(\frac{\mu_{\text{S}2}^{\text{bar}2}}{M_{\text{S}2}^2}\right)\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}M_{\text{S}2}^2\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{\text{S}1}^2 - M_{\text{S}2}^2)}
\end{aligned}$$

Weinberg sector

$$\frac{2\hbar M_F \left(M_F^2 \left(M_{\text{S}1}^2 \log\left(\frac{M_F^2}{M_{\text{S}1}^2}\right) - M_{\text{S}2}^2 \log\left(\frac{M_F^2}{M_{\text{S}2}^2}\right) \right) + M_{\text{S}1}^2 M_{\text{S}2}^2 \left(\log\left(\frac{M_F^2}{M_{\text{S}2}^2}\right) - \log\left(\frac{M_F^2}{M_{\text{S}1}^2}\right) \right) \right) (\lambda_{\text{T}3} H_{a_1} H_{a_3} \bar{\epsilon}_{a_1, a_2} \overline{y_{1, i_1, r_1}} \bar{\epsilon}_{a_3, a_4} \overline{y_{2, i_2, r_1}} \ell_{a_4, i_2}{}^T C P_L \ell_{a_2, i_1} + \overline{\lambda_{\text{T}3}} \epsilon_{a_1, a_3} y_{1, i_1, r_1} \epsilon_{a_2, a_4} H_{a_1}^\dagger H_{a_2}^\dagger y_{2, i_2, r_1} \overline{\ell_{a_3, i_1}} C P_R \overline{\ell_{a_4, i_2}}{}^T)}{3(M_F^2 - M_{\text{S}1}^2)(M_F^2 - M_{\text{S}2}^2)(M_{\text{S}1}^2 - M_{\text{S}2}^2)}$$

Weinberg coefficient C_5

$$\frac{2}{3} \left(- \frac{\hbar M_F^3 M_{\text{S}2}^2 \lambda_{\text{T}3} \log\left(\frac{M_F^2}{M_{\text{S}2}^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1}}{(M_F^2 - M_{\text{S}1}^2)(M_F^2 - M_{\text{S}2}^2)(M_{\text{S}1}^2 - M_{\text{S}2}^2)} + \frac{\hbar M_F^3 M_{\text{S}1}^2 \lambda_{\text{T}3} \log\left(\frac{M_F^2}{M_{\text{S}1}^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1}}{(M_F^2 - M_{\text{S}1}^2)(M_F^2 - M_{\text{S}2}^2)(M_{\text{S}1}^2 - M_{\text{S}2}^2)} - \frac{\hbar M_F M_{\text{S}1}^2 M_{\text{S}2}^2 \lambda_{\text{T}3} \log\left(\frac{M_F^2}{M_{\text{S}1}^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1}}{(M_F^2 - M_{\text{S}1}^2)(M_F^2 - M_{\text{S}2}^2)(M_{\text{S}1}^2 - M_{\text{S}2}^2)} + \frac{\hbar M_F M_{\text{S}1}^2 M_{\text{S}2}^2 \lambda_{\text{T}3} \log\left(\frac{M_F^2}{M_{\text{S}2}^2}\right) \overline{y_{1, i_1, r_1}} y_{2, i_2, r_1}}{(M_F^2 - M_{\text{S}1}^2)(M_F^2 - M_{\text{S}2}^2)(M_{\text{S}1}^2 - M_{\text{S}2}^2)} \right)$$

$$T3 - A, \alpha = 0 : \quad S_1 = (1, 0), \quad S_2 = (3, 1), \quad F = (2, \frac{1}{2})$$

Build T3 ingredients	Success Success	Matching Weinberg extraction	Success Success
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Accepted interactions

Yukawa1_GenericCG, Yukawa2_GenericCG, Scalar1SelfInv1, Scalar2SelfInv1, Scalar2SelfInv2, HiggsPortal1Inv1, HiggsPortal2Inv1, HiggsPortal2Inv2, ScalarCrossPortalInv1, T3ScalarMix_GenericCG.

BSM free Lagrangian $\mathcal{L}_{\text{free}}$

$$\begin{aligned} & i \overline{N_{j_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{j_1, r_1}) \\ & - M_F \overline{N_{j_1, r_1}} N_{j_1, r_1} \\ & + D_{\mu_1} (\text{NewScalar2}_{j_1}) D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger \\ & + D_{\mu_1} (\text{NewScalar1}) D_{\mu_1} (\text{NewScalar1})^\dagger \\ & - M_{S_2}^2 \text{NewScalar2}_{j_1} \text{NewScalar2}_{j_1}^\dagger \\ & - \text{NewScalar1} M_{S_1}^2 \text{NewScalar1}^\dagger \end{aligned}$$

Interaction Lagrangian $\mathcal{L}_{\text{int}}$

$$\begin{aligned} & \text{NewScalar1}^\dagger \overline{\text{T3Y1CG}_{a_1, j_1}} \overline{y_{1, i_1, r_1}} \ell_{a_1, i_1}^T C P_L N_{j_1, r_1} \\ & + \text{NewScalar1} \text{T3Y1CG}_{a_1, j_1} y_{1, i_1, r_1} \overline{\ell_{a_1, i_1}} C P_R \overline{N_{j_1, r_1}}^T \\ & + \text{NewScalar2}_{j_1} \overline{\text{T3Y2CG}_{a_1, j_1, j_1}} \overline{y_{2, i_1, r_1}} \overline{N_{j_1, r_1}} P_L \ell_{a_1, i_1} \\ & + \text{T3Y2CG}_{a_1, j_1, j_1} y_{2, i_1, r_1} \text{NewScalar2}_{j_1}^\dagger \overline{\ell_{a_1, i_1}} P_R N_{j_1, r_1} \\ & + \overline{\lambda_{T_3}} \text{NewScalar1}^\dagger \text{NewScalar2}_{j_1} H_{a_1}^\dagger H_{a_2}^\dagger \overline{\text{T3MixCG}_{a_1, a_2, j_1}} \\ & + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \overline{\text{T3Scalar2SelfCGInv1}_{j_1, j_3, j_2, j_4}} \\ & + \frac{1}{2} \text{lambdaS2Inv2} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \overline{\text{T3Scalar2SelfCGInv2}_{j_1, j_3, j_2, j_4}} \\ & + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \overline{\text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_1, j_2}} \\ & + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \overline{\text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_2, j_1}} \\ & + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \overline{\text{T3HiggsPortal2CGInv2}_{a_1, a_2, j_1, j_2}} \\ & + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \overline{\text{T3HiggsPortal2CGInv2}_{a_1, a_2, j_2, j_1}} \\ & + \text{NewScalar1} \lambda_{T_3} H_{a_1} H_{a_2} \text{NewScalar2}_{j_1}^\dagger \overline{\text{T3MixCG}_{a_1, a_2, j_1}} \\ & + \text{lambdaH1Inv1} \text{NewScalar1} \text{NewScalar1}^\dagger H_{a_2} \overline{\text{T3HiggsPortal1CGInv1}_{a_1, a_2} H_{a_1}^\dagger} \\ & + \text{lambdaI2Inv1} \text{NewScalar1} \text{NewScalar1}^\dagger \text{NewScalar2}_{j_2} \overline{\text{T3CrossScalarCGInv1}_{j_1, j_2} \text{NewScalar2}_{j_1}^\dagger} \\ & + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}^2 \left(\text{NewScalar1}^\dagger \right)^2 \end{aligned}$$

BSM-induced EFT contribution $\Delta\mathcal{L}_{\text{EFT}}$

$$\begin{aligned} & - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S_1}^2)^2} \\ & - \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S_2}^2)^2} \\ & - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S_1}^2)^2} \end{aligned}$$

$$\begin{aligned}
& - \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{2 \hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2 \hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2 \hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2 \hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{1}{4} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{2\hbar (W_{\mu_1\mu_2}^{A_1})^2}{3\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& - \frac{1}{6} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& + \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{3} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \sqrt{3} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\sqrt{3} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\sqrt{3} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\sqrt{3} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\sqrt{3} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S2}^2)^2} \\
& + \text{CH21} H_{a_1} H_{a_1}^\dagger \\
& + \hbar C_{H2} H_{a_2} \text{T3HiggsPortal1CGInv1}_{a_1,a_2} H_{a_1}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_1}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_1}^2)} \\
& - \frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_2}^2)} \\
& - \frac{\sqrt{3} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_2}^2)} \\
& - \frac{\sqrt{3} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\sqrt{3} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{3 \text{lambdaH2Inv}^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \text{lambdaH2Inv}^2 \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH1Inv}^2 \hbar H_{a_3} H_{a_4} \text{T3HiggsPortal1CGInv1}_{a_1,a_3} \text{T3HiggsPortal1CGInv1}_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \text{lambdaH1Inv}^2 \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) H_{a_3} H_{a_4} \text{T3HiggsPortal1CGInv1}_{a_1,a_3} \text{T3HiggsPortal1CGInv1}_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{2 \hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} \epsilon} \\
& + \frac{2 \hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2 \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2 \hbar \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2 \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) \overline{\lambda_{\text{T3}}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{\text{T3}} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} (M_{S_1}^2 - M_{S_2}^2)}
\end{aligned}$$

Weinberg sector

$$\begin{aligned}
& - 2 \hbar M_F \left(M_F^2 \left(M_{S_1}^2 \log\left(\frac{M_F^2}{M_{S_1}^2}\right) - M_{S_2}^2 \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \right) + M_{S_1}^2 M_{S_2}^2 \left(\log\left(\frac{M_F^2}{M_{S_1}^2}\right) - \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \right) \right) \left(\lambda_{\text{T3}} H_{a_1} H_{a_3} \overline{e_{a_1,a_2}} \overline{y_{1,i_1,r_1}} \overline{e_{a_3,a_4}} \overline{y_{2,i_2,r_1}} \overline{e_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} + \overline{\lambda_{\text{T3}}} \epsilon_{a_1,a_3} y_{1,i_1,r_1} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger y_{2,i_2,r_1} \overline{\ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T \right) \\
& - \frac{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)}{
\end{aligned}$$

Weinberg coefficient C_5

$$-\frac{2}{3} \left(-\frac{\sqrt{3}\hbar M_F^3 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{3}\hbar M_F^3 M_{S1}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} - \frac{\sqrt{3}\hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{3}\hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \right)$$

$$T3 - D, \alpha = -2: \quad S_1 = (3, -1), \quad S_2 = (1, 0), \quad F = (2, -\frac{1}{2})$$

Build T3 ingredients	Success Success	Matching Weinberg extraction	Success Success
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Accepted interactions

Yukawa1_GenericCG, Yukawa2_GenericCG, Scalar1SelfInv1, Scalar1SelfInv2, Scalar2SelfInv1, HiggsPortal1Inv1, HiggsPortal1Inv2, HiggsPortal2Inv1, ScalarCrossPortalInv1, T3ScalarMix_GenericCG.

BSM free Lagrangian $\mathcal{L}_{\text{free}}$

$$\begin{aligned} & i\overline{N_{j_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{j_1, r_1}) \\ & - M_F \overline{N_{j_1, r_1}} N_{j_1, r_1} \\ & + D_{\mu_1} (\text{NewScalar1}_{j_1}) D_{\mu_1} (\text{NewScalar1}_{j_1})^\dagger \\ & + D_{\mu_1} (\text{NewScalar2}) D_{\mu_1} (\text{NewScalar2})^\dagger \\ & + M_{S1}^2 (-\text{NewScalar1}_{j_1}) \text{NewScalar1}_{j_1}^\dagger \\ & - \text{NewScalar2} M_{S2}^2 \text{NewScalar2}^\dagger \end{aligned}$$

Interaction Lagrangian $\mathcal{L}_{\text{int}}$

$$\begin{aligned} & \text{NewScalar1}_{j_1}^\dagger \overline{\text{T3Y1CG}_{a_1, j_1, j_1}} \overline{y_{1, i_1, r_1}} \ell_{a_1, i_1}^T C P_L N_{j_1, r_1} \\ & + \text{NewScalar1}_{j_1} \overline{\text{T3Y1CG}_{a_1, j_1, j_1}} y_{1, i_1, r_1} \overline{\ell_{a_1, i_1}} C P_R \overline{N_{j_1, r_1}}^T \\ & + \text{NewScalar2} \overline{\text{T3Y2CG}_{a_1, j_1}} \overline{y_{2, i_1, r_1}} \overline{N_{j_1, r_1}} P_L \ell_{a_1, i_1} \\ & + \text{NewScalar2}^\dagger \overline{\text{T3Y2CG}_{a_1, j_1}} y_{2, i_1, r_1} \overline{\ell_{a_1, i_1}} P_R N_{j_1, r_1} \\ & + \text{NewScalar2} \overline{\lambda_{T3}} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger H_{a_2}^\dagger \overline{\text{T3MixCG}_{a_1, a_2, j_1}} \\ & + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \overline{\text{T3Scalar1SelfCGInv1}_{j_1, j_3, j_2, j_4}} \\ & + \frac{1}{2} \text{lambdaS1Inv2} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \overline{\text{T3Scalar1SelfCGInv2}_{j_1, j_3, j_2, j_4}} \\ & + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \overline{\text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_1, j_2}} \\ & + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \overline{\text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_2, j_1}} \\ & + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \overline{\text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_1, j_2}} \\ & + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \overline{\text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_2, j_1}} \\ & + \lambda_{T3} \text{NewScalar2}^\dagger H_{a_1} \text{NewScalar1}_{j_1} H_{a_2} \overline{\text{T3MixCG}_{a_1, a_2, j_1}} \\ & + \text{lambdaH2Inv1} \text{NewScalar2} \text{NewScalar2}^\dagger H_{a_2} \overline{\text{T3HiggsPortal2CGInv1}_{a_1, a_2}} H_{a_1}^\dagger \\ & + \text{lambdaI2Inv1} \text{NewScalar2} \text{NewScalar2}^\dagger \text{NewScalar1}_{j_2} \overline{\text{T3CrossScalarCGInv1}_{j_1, j_2}} \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}^2 \left(\text{NewScalar2}^\dagger \right)^2 \end{aligned}$$

BSM-induced EFT contribution $\Delta\mathcal{L}_{\text{EFT}}$

$$\begin{aligned} & - \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{1, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S1}^2)^2} \\ & - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{2, i_2, r_1} Y_{e, i_3, i_1} M_F^4}{4 (M_F^2 - M_{S2}^2)^2} \\ & - \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{1, i_3, r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S1}^2)^2} \end{aligned}$$

$$\begin{aligned}
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{\sqrt{3} (M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3} \hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\sqrt{3} \hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{3} \hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)} \\
& + \frac{\sqrt{3} \hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4 (M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2 (M_F^2 - M_{S_2}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S1}^2 M_{S2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& + \frac{2\hbar \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{2\hbar \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S1}^2 M_{S2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \\
& - \frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{1}{4} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{2\hbar (W_{\mu_1\mu_2}^{A_1})^2}{3\epsilon} \\
& - \frac{1}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& - \frac{1}{6} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& + \frac{\sqrt{3} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \sqrt{3} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\sqrt{3} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\sqrt{3} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S1}^2)} \\
& - \frac{\sqrt{3} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& - \frac{\sqrt{3} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S1}^2)^2} \\
& + \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S2}^2)} \\
& - \frac{\hbar \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S2}^2)^2} \\
& + C_{H2} H_{a_1} H_{a_1}^\dagger \\
& + CH21 \hbar H_{a_2} T3\text{HiggsPortal}2\text{CGInv}1_{a_1,a_2} H_{a_1}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \frac{\sqrt{3}\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \sqrt{3}\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\sqrt{3}\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_1}^2)} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_1}^2)} \\
& - \frac{\sqrt{3}\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_2}^2)} \\
& - \frac{\hbar \log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_2}^2)} \\
& - \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\hbar \log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{3\text{lambdaH1Inv}2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv}2^2 \hbar \log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\text{lambdaH2Inv}1^2 \hbar H_{a_3} H_{a_4} \text{T3HiggsPortal2CGInv}1_{a_1,a_3} \text{T3HiggsPortal2CGInv}1_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \text{lambdaH2Inv}1^2 \hbar \log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right) H_{a_3} H_{a_4} \text{T3HiggsPortal2CGInv}1_{a_1,a_3} \text{T3HiggsPortal2CGInv}1_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon} \\
& + \frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\hbar \log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\hbar \log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S_1}^2 - M_{S_2}^2)}
\end{aligned}$$

Weinberg sector

$$\begin{aligned}
& - 2\hbar M_F \left(M_F^2 \left(M_{S_1}^2 \log\left(\frac{M_{S_1}^2}{M_{S_1}^2}\right) - M_{S_2}^2 \log\left(\frac{M_{S_2}^2}{M_{S_2}^2}\right) \right) + M_{S_1}^2 M_{S_2}^2 \left(\log\left(\frac{M_{S_1}^2}{M_{S_1}^2}\right) - \log\left(\frac{M_{S_2}^2}{M_{S_2}^2}\right) \right) \right) \left(\lambda_{T3} H_{a_1} H_{a_3} \overline{e_{a_1,a_2} y_{1,i_1,r_1} e_{a_3,a_4} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} + \overline{\lambda_{T3} e_{a_1,a_3} y_{1,i_1,r_1} e_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger y_{2,i_2,r_1} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T \right) \\
& - \frac{2\hbar M_F \left(M_F^2 \left(M_{S_1}^2 \log\left(\frac{M_{S_1}^2}{M_{S_1}^2}\right) - M_{S_2}^2 \log\left(\frac{M_{S_2}^2}{M_{S_2}^2}\right) \right) + M_{S_1}^2 M_{S_2}^2 \left(\log\left(\frac{M_{S_1}^2}{M_{S_1}^2}\right) - \log\left(\frac{M_{S_2}^2}{M_{S_2}^2}\right) \right) \right)}{\sqrt{3}(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}
\end{aligned}$$

Weinberg coefficient C_5

$$-\frac{2}{3} \left(-\frac{\sqrt{3}\hbar M_F^3 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{3}\hbar M_F^3 M_{S1}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} - \frac{\sqrt{3}\hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{3}\hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i1,r1} y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \right)$$

$$T3 - E, \alpha = 0 : \quad S_1 = (3, 0), \quad S_2 = (3, 1), \quad F = (2, \frac{1}{2})$$

Build T3 ingredients	Success Success	Matching Weinberg extraction	Success Success
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Accepted interactions

Yukawa1_GenericCG, Yukawa2_GenericCG, Scalar1SelfInv1, Scalar1SelfInv2, Scalar2SelfInv1, Scalar2SelfInv2, HiggsPortal1Inv1, HiggsPortal1Inv2, HiggsPortal2Inv1, HiggsPortal2Inv2, ScalarCrossPortalInv1, ScalarCrossPortalInv2, ScalarCrossPortalInv3, T3ScalarMix_GenericCG.

BSM free Lagrangian $\mathcal{L}_{\text{free}}$

$$\begin{aligned} & \overline{iN_{j_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{j_1, r_1}) \\ & - M_F \overline{N_{j_1, r_1}} N_{j_1, r_1} \\ & + D_{\mu_1} (\text{NewScalar1}_{j_1}) D_{\mu_1} (\text{NewScalar1}_{j_1})^\dagger \\ & + D_{\mu_1} (\text{NewScalar2}_{j_1}) D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger \\ & + M_{S1}^2 (-\text{NewScalar1}_{j_1}) \text{NewScalar1}_{j_1}^\dagger \\ & - M_{S2}^2 \text{NewScalar2}_{j_1} \text{NewScalar2}_{j_1}^\dagger \end{aligned}$$

Interaction Lagrangian $\mathcal{L}_{\text{int}}$

$$\begin{aligned} & \overline{\ell_{a_1, i_1}} C P_R \overline{N_{j_1, r_1}}^T \text{NewScalar1}_{j_1} \text{T3Y1CG}_{a_1, j_1, j_1} y_{1, i_1, r_1} \\ & + \overline{y_{2, i_1, r_1}} \overline{N_{j_1, r_1}} P_L \ell_{a_1, i_1} \text{NewScalar2}_{j_1} \overline{\text{T3Y2CG}}_{a_1, j_1, j_1} \\ & + \overline{y_{1, i_1, r_1}} \ell_{a_1, i_1}^T C P_L N_{j_1, r_1} \overline{\text{T3Y1CG}}_{a_1, j_1, j_1} \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \overline{\lambda_{\text{T3}}} \text{NewScalar2}_{j_2} \overline{\text{T3MixCG}}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\ & + \frac{1}{2} \text{lambdaS1Inv2} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\ & + \overline{\ell_{a_1, i_1}} P_R N_{j_1, r_1} \text{T3Y2CG}_{a_1, j_1, j_1} y_{2, i_1, r_1} \text{NewScalar2}_{j_1}^\dagger \\ & + H_{a_1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3MixCG}_{a_1, a_2, j_2, j_1} \lambda_{\text{T3}} \text{NewScalar2}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambda12Inv1} \text{NewScalar1}_{j_3} \text{NewScalar2}_{j_4} \text{T3CrossScalarCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \text{lambda12Inv2} \text{NewScalar1}_{j_3} \text{NewScalar2}_{j_4} \text{T3CrossScalarCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \text{lambda12Inv3} \text{NewScalar1}_{j_3} \text{NewScalar2}_{j_4} \text{T3CrossScalarCGInv3}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \frac{1}{2} \text{lambdaS2Inv2} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \end{aligned}$$

BSM-induced EFT contribution $\Delta\mathcal{L}_{\text{EFT}}$

$$\begin{aligned}
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{4(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4}{4(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{4(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} \ell_{a_4,i_2}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} \ell_{a_4,i_2}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \lambda_{T3} \ell_{a_3,i_1} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \lambda_{T3} \ell_{a_3,i_1} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_1}^2)} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_1}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{4(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_1}^2)} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_1}^2)} \\
& + \frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{4(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\sqrt{3}\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{3}\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{4(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\
& - \frac{1}{2}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{1}{4}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) B_{\mu_1\mu_2}^2 \\
& - \frac{5\hbar (W_{\mu_1\mu_2}^{A_1})^2}{6\epsilon} \\
& - \frac{1}{2}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& - \frac{1}{6}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& - \frac{1}{6}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& + \frac{\sqrt{3}\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8}\sqrt{3}\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\sqrt{3}\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S_1}^2)} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S_1}^2)} \\
& - \frac{\sqrt{3}\hbar y_{1,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\sqrt{3}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3}\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon} \\
& - \frac{1}{8}\sqrt{3}\hbar y_{2,i_3,r_1} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}
\end{aligned}$$

$$\begin{aligned}
& - \frac{\sqrt{3\hbar y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S_2}^2)} \\
& - \frac{\sqrt{3\hbar} \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S_2}^2)} \\
& - \frac{\sqrt{3\hbar y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\sqrt{3\hbar} \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) \overline{y_{2,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^4 y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4(M_F^2 - M_{S_2}^2)^2} \\
& + C_{H_2} H_{a_1} H_{a_1}^\dagger \\
& + \frac{\sqrt{3\hbar y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \sqrt{3\hbar y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\sqrt{3\hbar y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_1}^2)} \\
& - \frac{\sqrt{3\hbar} \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_1}^2)} \\
& - \frac{\sqrt{3\hbar y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\sqrt{3\hbar} \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^4 y_{1,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{3\hbar y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon} \\
& - \frac{1}{8} \sqrt{3\hbar y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \\
& - \frac{\sqrt{3\hbar y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_2}^2)} \\
& - \frac{\sqrt{3\hbar} \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger}{2(M_F^2 - M_{S_2}^2)} \\
& - \frac{\sqrt{3\hbar y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\sqrt{3\hbar} \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^4 y_{2,i_3,r_1} H_{a_1}^\dagger}{4(M_F^2 - M_{S_2}^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{3\text{lambdaH1Inv} 2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3\text{lambdaH2Inv} 2^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv} 2^2 \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{3}{2} \text{lambdaH2Inv} 2^2 \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\
& - \frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\
& - \frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S_2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2}
\end{aligned}$$

Weinberg sector

$$\frac{2\sqrt{\frac{2}{3}}\hbar M_F \left(M_F^2 \left(M_{S_1}^2 \log\left(\frac{M_F^2}{M_{S_1}^2}\right) - M_{S_2}^2 \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \right) + M_{S_1}^2 M_{S_2}^2 \left(\log\left(\frac{M_F^2}{M_{S_2}^2}\right) - \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \right) \right) (\lambda_{T3} H_{a_1} H_{a_3} \epsilon_{a_1,a_2} \overline{y_{1,i_1,r_1}} \epsilon_{a_3,a_4} \overline{y_{2,i_2,r_1}} \ell_{a_4,i_2}{}^T C P_L \ell_{a_2,i_1} + \overline{\lambda_{T3}} \epsilon_{a_1,a_3} y_{1,i_1,r_1} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger y_{2,i_2,r_1} \overline{\ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}{}^T)}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)}$$

Weinberg coefficient C_5

$$\frac{2}{3} \left(- \frac{\sqrt{6}\hbar M_F^3 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} + \frac{\sqrt{6}\hbar M_F^3 M_{S_1}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} - \frac{\sqrt{6}\hbar M_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} + \frac{\sqrt{6}\hbar M_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1}}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \right)$$